

Early Detection of Cardiac Arrest in Newborns Using Statistical Machine Learning in Cardiac ICU



Introduction

Cardiac arrest in Newborns is a life-threatening condition that can cause serious complications or death. Early detection is crucial for ensuring timely treatment and better outcomes. Key signs include a rapid heart rate, difficulty breathing, bluish skin (cyanosis), unresponsiveness, and reduced movement. These symptoms require immediate medical attention.

Several risk factors can increase the likelihood of cardiac arrest in Newborns, such as low birth weight, preterm birth, difficult delivery, a family history of cardiac issues, and maternal high blood pressure during pregnancy. Reviewing the baby's medical history is also important to assess potential risks.

To detect cardiac arrest early, regular monitoring of heart rate and respiratory rate is essential. Pulse oximetry, a non-invasive and painless method, is commonly used to measure the oxygen level in the baby's blood, helping identify early signs of distress and preventing serious outcomes.

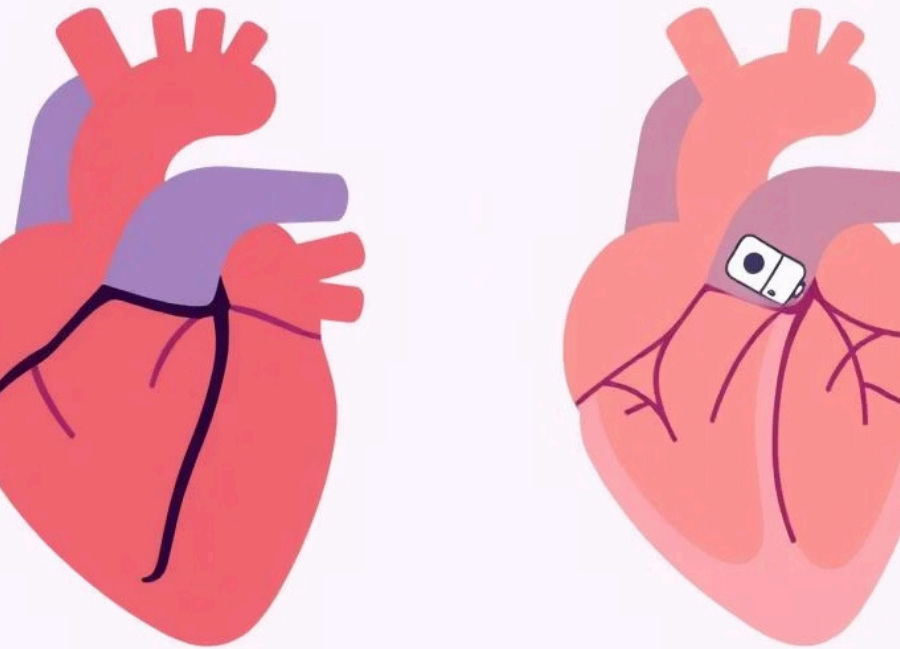
Literature Review

Previous research has shown the potential of machine learning in predicting critical conditions in neonatal ICUs. Studies using models like logistic regression, decision trees, and LSTM networks have demonstrated success in detecting early signs of cardiac arrest using physiological data. Traditional threshold-based systems often generate false alarms, causing alarm fatigue. Machine learning offers improved accuracy and early warning capabilities. However, challenges like limited neonatal-specific datasets and the need for real-time implementation remain. Overall, literature supports the use of ML for enhancing early detection in newborn cardiac care.



Problem Statement

Cardiac arrest in newborns is a life-threatening emergency requiring immediate action. However, early signs are often subtle and missed by traditional monitoring systems, leading to delayed interventions. High false alarm rates further strain ICU staff and reduce response efficiency. This project aims to use statistical machine learning to analyze real-time physiological data (e.g., heart rate, oxygen levels) to detect early warning signs of cardiac arrest. The goal is to enable timely alerts, improve survival rates, and enhance neonatal care through predictive, data-driven insights.



Existing Systems

Carlisle et al. [21] discusses heart failure, a condition where the heart can't pump enough blood due to causes like high blood pressure, coronary artery disease, and diabetes. Atrial fibrillation, an irregular heartbeat, is a common cause, leading to symptoms such as shortness of breath and fatigue. Treatment typically includes medications, lifestyle changes, and sometimes surgery.



Disadvantages

- *Data Complexity:* Existing models struggle to interpret large and complex datasets for detecting cardiac arrest in newborns.
- *Data Availability:* Insufficient data can reduce model accuracy and effectiveness.
- *Incorrect Labeling:* Models depend on accurate data labeling; incorrect labels lead to inaccurate predictions.

Proposed System

The proposed system uses machine learning to detect cardiac arrest in newborns early by analyzing vital signs like heart rate, breathing, and oxygen levels. ML models can identify subtle changes that may not be visible to the naked eye. These models predict risk factors and provide alerts hours before traditional methods. This allows timely medical intervention, increasing survival chances and reducing complications. The system also helps identify high-risk infants for closer monitoring. ML enables continuous, non-invasive monitoring and personalized care based on patient data. Overall, it improves detection accuracy and supports better neonatal health outcomes.



Advantages

- *Personalized Care:* Customizes monitoring based on patient data.
- *Early Detection:* Predicts cardiac arrest ahead of traditional methods.
- *Improved Accuracy:* Identifies subtle vital sign changes.
- *Increased Survival:* Enables timely intervention, reducing complications.
- *Risk Identification:* Detects high-risk infants for closer monitoring.

Software Requirements

- Operating system : Windows 7 Ultimate.
- Coding Language : Python.
- Front-End : Python.
- Back-End : Django-ORM
- Designing : Html, CSS, JavaScript.
- Data Base : MySQL (WAMP Server).





Hardware Requirements

1. Processor - Pentium –IV
2. RAM - 4 GB (min)
3. Hard Disk - 20 GB
4. Network - stable internet connection



Methodology

Future Scope

- *Larger Datasets:* Use more neonatal data for better prediction accuracy.
- *Real-time Integration:* Integrate models with ICU systems for continuous monitoring.
- *Wearable Devices:* Enable remote monitoring for high-risk infants.
- *Clinical Validation:* Validate models across multiple hospitals for real-world effectiveness.



conclusion

The proposed machine learning model accurately detects early signs of cardiac arrest in newborns using vital sign data. It enables timely intervention, improving survival rates and reducing CICU stays. Future enhancements include real-time data use and AI-driven personalized care.

Reference

- 1] E. Choi, A. Schuetz, W. F. Stewart, and J. Sun, “Using recurrent neural network models for early detection of heart failure onset,” J. Amer. Med. Inform. Assoc., vol. 24, no. 2, pp. 361–370, Mar. 2017.
- 2] K.W. Johnson, J. T. Soto, B. S. Glicksberg, K. Shameer, R. Miotto, M. Ali, E. Ashley, and J. T. Dudley, “Artificial intelligence in cardiology,” J. Amer. College Cardiol., vol. 71, pp. 2668–2679, Jun. 2018.